RAB TECH ACADEMY
DATA SCIENCE INTERNSHIP

EXPLAINABLE MACHINE LEARNING FOR DIGITAL PAYMENT FRAUD DETECTION

A research capstone on imbalanced transaction classification, model comparison and feature importance

Research question
Can machine learning accurately detect fraudulent digital payment transactions and identify the key factors associated with fraud risk?

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Project: Task 6 – Research Capstone & Whitepaper
Dataset: Credit Card Fraud Detection (Kaggle)

Research report prepared from the submitted Jupyter Notebook and its recorded results.

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Note: The dataset is a historical credit-card transaction dataset. It is used here as a proxy for digital-payment fraud analysis; it is not specifically a UPI dataset.

1. Abstract

Fraud detection is a challenging classification problem because fraudulent transactions are rare compared with legitimate transactions. This study evaluates two supervised machine-learning approaches—Logistic Regression and Random Forest—on the Kaggle Credit Card Fraud Detection dataset. The uploaded notebook reports 284,807 transactions with 492 fraud cases and no missing values. An 80/20 stratified train-test split was used, with 98 fraud cases in the test set. Performance was assessed using precision, recall, F1-score, Precision–Recall Area Under the Curve (PR-AUC), and ROC-AUC. Logistic Regression achieved 0.71897 PR-AUC and 0.92 fraud recall, but generated many false positives. Random Forest achieved 0.85420 PR-AUC, 0.96 fraud precision, 0.74 fraud recall, and an F1-score of 0.84. The Random Forest feature-importance output ranked V14, V10, V12, V4, and V17 highest. These results suggest that the Random Forest model provides a stronger precision–recall balance in this experiment, while the final operating threshold should depend on the relative cost of missed fraud and unnecessary alerts.

Keywords: fraud detection, imbalanced classification, Logistic Regression, Random Forest, PR-AUC, explainable machine learning.

2. Introduction and Research Question

Digital transactions offer speed and convenience, but fraudulent activity can cause financial loss for customers and institutions. Automated fraud detection systems must identify suspicious transactions among a much larger volume of legitimate activity. This class imbalance makes accuracy alone an inadequate measure: a model can appear highly accurate while failing to identify rare fraud cases.

This project compares two commonly used classifiers and examines feature importance from the Random Forest model. The purpose is not to claim a production-ready fraud system, but to conduct a reproducible experiment, report appropriate evaluation metrics, and identify limitations that should guide further work.

Research question: Can machine learning accurately detect fraudulent digital payment transactions and identify the key factors associated with fraud risk?

3. Background and Related Work

Random Forest combines multiple decision trees to capture nonlinear patterns and interactions. Logistic Regression provides a comparatively simple baseline whose output can be interpreted as a linear relationship between predictors and log-odds. Both can be useful, but model quality depends on the data, class imbalance, feature representation, and decision threshold.

Precision–Recall analysis is especially informative when the positive class is rare. Precision describes how many flagged transactions are actually fraudulent; recall describes how many fraud cases are found. Increasing recall may increase false alerts, while prioritizing precision may leave more fraud undetected. These trade-offs are central to operational fraud monitoring.

The dataset contains anonymized variables V1–V28, which were produced through a transformation in the source data. Their feature-importance values should therefore be treated as model-specific predictive signals, not direct explanations such as customer identity, merchant type, or transaction intent.

4. Dataset Description and Exploratory Analysis

The notebook loads the Kaggle Credit Card Fraud Detection dataset. It contains 284,807 rows and 31 columns: Time, V1–V28, Amount, and Class. The target variable Class is 0 for genuine transactions and 1 for fraudulent transactions. The notebook reports zero missing values.

Dataset measure	Notebook result
Total transactions	284,807
Genuine transactions (Class = 0)	284,315
Fraud transactions (Class = 1)	492

Dataset measure	Notebook result
Fraud proportion	0.1727%
Missing values	0
Features used in modelling	30 (all columns except Class)

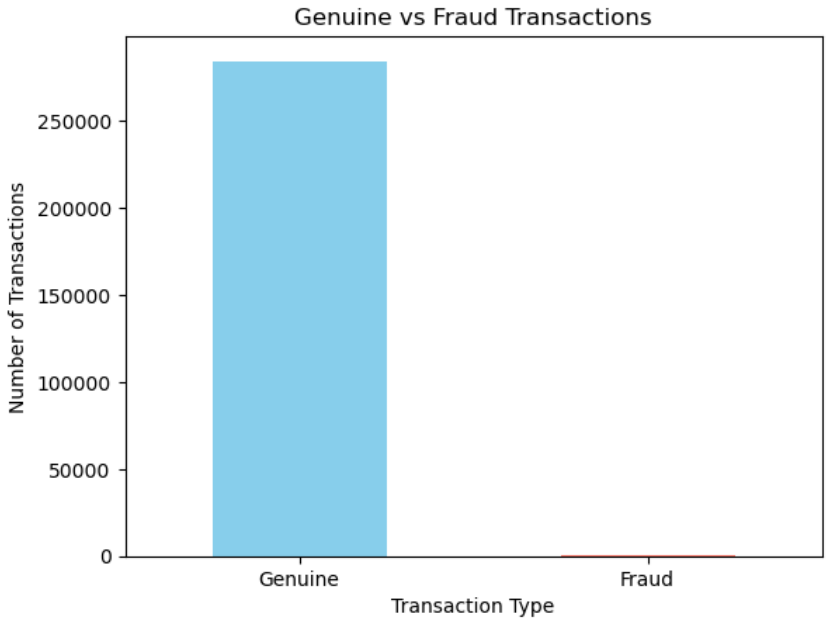


Figure 1. Class distribution in the dataset. Fraud cases form a very small minority of all transactions.

The class distribution shows a severe imbalance. The modelling workflow therefore uses stratified splitting and class weighting for both classifiers, helping preserve the fraud proportion in train and test partitions and making fraud detection a stronger training objective.

Transaction Amounts

The notebook reports a mean transaction amount of 88.29 and median of 22.00 for genuine transactions. Fraudulent transactions have a mean amount of 122.21 and median of 9.25. These summaries show that the average and median differ substantially, and that amount alone is unlikely to separate all fraud from genuine activity.

Amount statistic	Genuine	Fraud
Mean	88.291	122.211
Median	22.00	9.25
Maximum	25,691.16	2,125.87

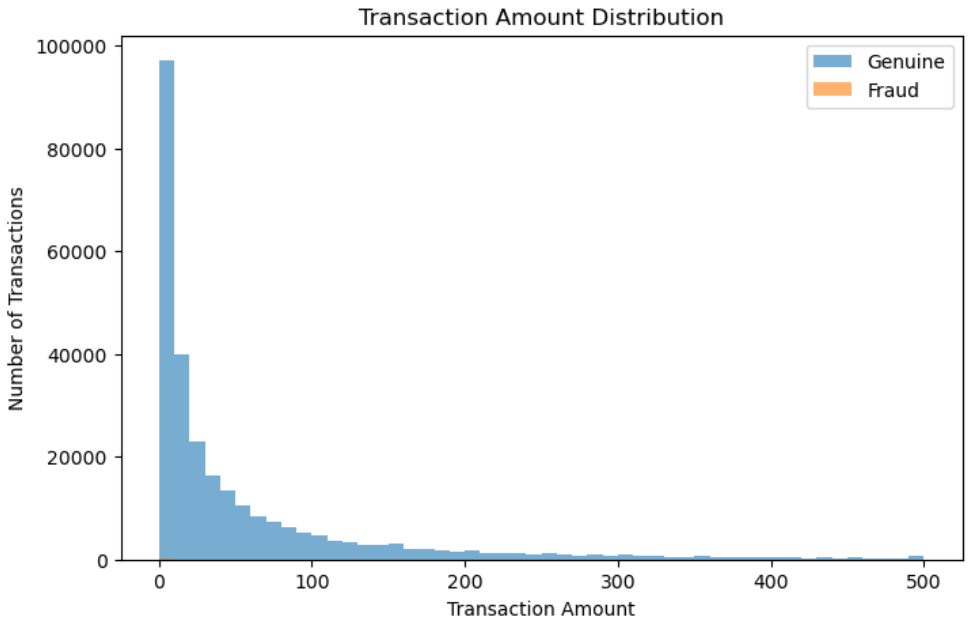


Figure 2. Transaction amount distribution by class, as plotted in the submitted notebook.

5. Methodology

Data preparation and split

The target column Class was separated from the predictor matrix. The data was divided into 80% training and 20% testing subsets using a stratified split and random_state=42. The notebook reports 227,845 training rows and 56,962 test rows; the test partition contains 98 fraud cases.

Model 1: Logistic Regression

The notebook implements Logistic Regression in a scikit-learn Pipeline with StandardScaler and LogisticRegression(class_weight='balanced', max_iter=1000, random_state=42). Scaling is included in the pipeline, and class weighting is used to account for the minority fraud class.

Model 2: Random Forest

The notebook uses RandomForestClassifier with 100 trees, class_weight='balanced', random_state=42, and n_jobs=-1. Random Forest can model nonlinear decision boundaries and interactions between input variables.

Evaluation metrics

Precision, recall, F1-score, PR-AUC (Average Precision), ROC-AUC, and confusion matrices were recorded. PR-AUC is emphasized because fraud represents only 0.1727% of the dataset. Results are from the notebook's particular split and settings; they are not a guarantee of future or real-world performance.

6. Results and Model Comparison

Metric (fraud class unless noted)	Logistic Regression	Random Forest
Precision	0.06	0.96
Recall	0.92	0.74
F1-score	0.11	0.84
PR-AUC / Average Precision	0.71897	0.85420
ROC-AUC	0.97208	0.95291

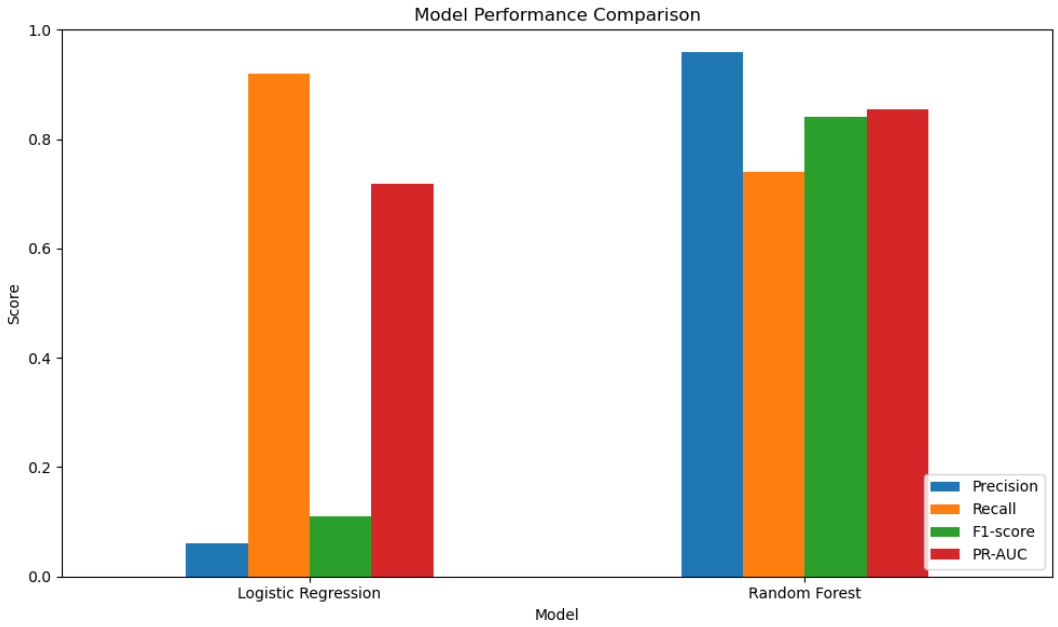


Figure 3. Model performance comparison. The plotted measures are those recorded in the notebook.

Random Forest achieved the higher PR-AUC (0.85420 compared with 0.71897) and substantially higher fraud precision and F1-score. Logistic Regression achieved higher fraud recall (0.92 compared with 0.74) and a higher ROC-AUC. This demonstrates that the preferred model depends on the operational objective: finding as many fraud cases as possible versus reducing false alerts.

Main result: In this experiment, Random Forest provides the stronger PR-AUC and F1-score, while Logistic Regression catches a larger proportion of fraud cases but flags many more genuine transactions.

7. Confusion Matrix and Error Analysis

The test set contains 56,864 genuine transactions and 98 fraud transactions for the Random Forest evaluation. Its confusion matrix is [[56861, 3], [25, 73]], using the class order genuine then fraud. This corresponds to 56,861 correctly identified genuine transactions, 3 genuine transactions incorrectly flagged as fraud, 73 fraud cases detected, and 25 fraud cases missed.

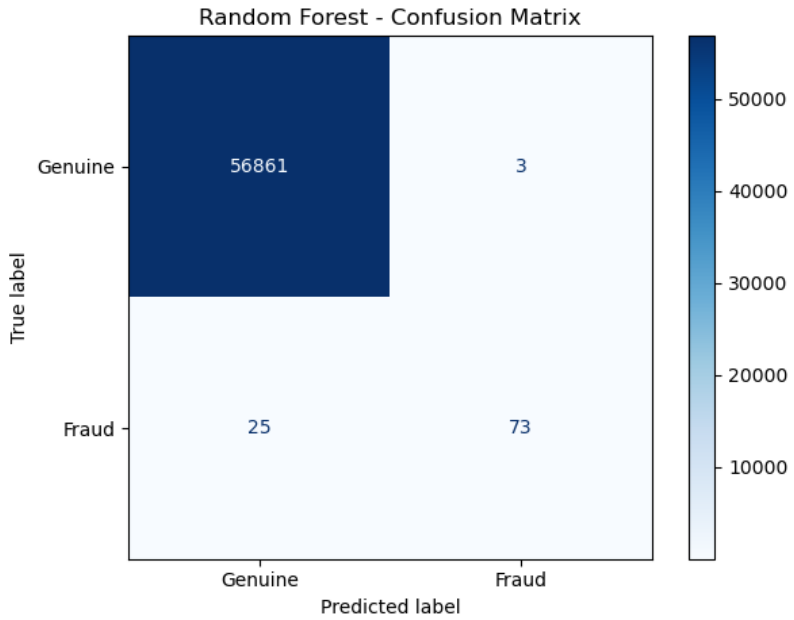


Figure 4. Random Forest confusion matrix from the submitted notebook.

Random Forest outcome	Count	Interpretation
True negatives	56,861	Genuine transactions correctly classified
False positives	3	Genuine transactions flagged as fraud
False negatives	25	Fraud transactions missed
True positives	73	Fraud transactions correctly detected

Logistic Regression produced the confusion matrix [[55478, 1386], [8, 90]]. It detected 90 of the 98 test fraud cases, but also flagged 1,386 genuine transactions as fraud. The contrast illustrates why a single metric is not enough to judge a fraud classifier.

8. Feature Importance and Interpretability

The notebook extracts feature_importances_ from the fitted Random Forest model and ranks the variables. The five highest-ranked variables are V14 (0.179857), V10 (0.115442), V12 (0.096206), V4 (0.095646), and V17 (0.095113).

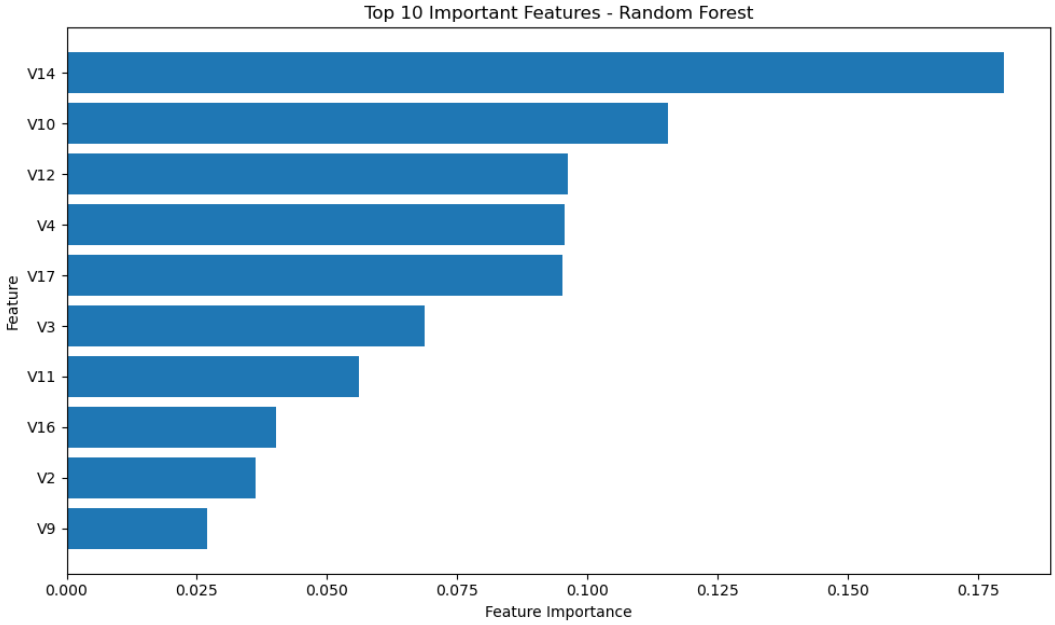


Figure 5. Top features ranked by Random Forest feature importance.

Rank	Feature	Importance
1	V14	0.179857
2	V10	0.115442
3	V12	0.096206
4	V4	0.095646
5	V17	0.095113
6	V3	0.068669
7	V11	0.056112
8	V16	0.040303
9	V2	0.036265

Rank	Feature	Importance
10	V9	0.026888

Feature importance describes how the fitted Random Forest used the variables to make predictions. It does not establish causation. Since V1–V28 are anonymized transformed variables, this analysis cannot directly tell an investigator what real-world customer or merchant behaviour caused a transaction to be fraudulent. Further explanation methods, such as SHAP, could provide case-level insight but would still need careful interpretation.

9. Discussion and Practical Implications

The results suggest that class-aware training and evaluation metrics are essential for rare-event detection. Random Forest appears preferable if the goal is to obtain a better balance of precision and recall under the reported test setup. Logistic Regression may be useful where high recall is a priority and the operational team can review a larger number of alerts.

A real fraud-monitoring workflow should not automatically block transactions solely on this experiment. A practical system would typically combine model scores with a calibrated decision threshold, human review for uncertain cases, customer-protection procedures, monitoring for data drift, and a process for investigating false positives and false negatives.

The dataset is historical European credit-card data. Although this project frames the topic as digital-payment fraud detection, its results should not be described as validated UPI-specific performance. A UPI deployment would require representative UPI transaction data, appropriate governance, and validation with domain experts.

10. Limitations and Future Work

• The dataset is highly imbalanced and contains only 492 fraud cases in total; the test partition contains 98.

• Results depend on one stratified train-test split and the chosen model settings. Cross-validation and repeated evaluation would provide a more robust estimate.

• The notebook uses default prediction thresholds; threshold tuning could change the precision–recall trade-off.

• Random Forest feature importance is not a causal explanation and may be affected by feature correlations.

• V1–V28 are anonymized features, limiting business-level interpretability.

• The dataset is not specifically UPI data, so findings should not be generalized to UPI without new validation.

• Future work could add cross-validation, calibration, threshold analysis, SHAP explanations, time-based validation, and monitoring for data drift.

11. Conclusion

This research capstone compared Logistic Regression and Random Forest for detecting rare fraudulent transactions using the submitted notebook's Kaggle credit-card dataset. Random Forest achieved the higher PR-AUC (0.85420) and F1-score (0.84), while Logistic Regression achieved higher fraud recall (0.92) but substantially more false positives. The Random Forest ranking identified V14, V10, V12, V4, and V17 as the most influential features within this fitted model. The findings demonstrate the importance of class imbalance, appropriate evaluation metrics, and careful interpretation of model explanations. Further validation on representative, domain-specific transactions is required before real-world use.

12. References

• Breiman, L. (2001). Random Forests. *Machine Learning, 45*, 5–32. https://doi.org/10.1023/A:1010933404324

• Saito, T., & Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLOS ONE, 10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

• Kaggle. Credit Card Fraud Detection dataset (mlg-ulb/creditcardfraud). https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

• Scikit-learn documentation. Logistic Regression, Random Forest Classifier, classification metrics, and ConfusionMatrixDisplay. https://scikit-learn.org/stable/

Appendix: Reproducibility Notes

The report was drafted from the submitted notebook outputs. The notebook uses train_test_split with test_size=0.2, random_state=42, and stratify=y. Model settings are listed in the Methodology section. To make the project reproducible on GitHub, replace the personal Windows CSV path with a relative path or a short data-download instruction, and do not commit the raw dataset if its size or licence makes that inappropriate.

Dataset source and evaluation results are reported as present in the notebook; no claim is made that the model has been deployed or validated in a live payment system.